```
ERROR 1050 (42S01): Table 'employees' already exists
mysql> CREATE TABLE employees (
    ->     employee_id INT PRIMARY KEY,
    ->     first_name VARCHAR(50),
    ->     last_name VARCHAR(50),
    ->     email VARCHAR(100),
    ->     department VARCHAR(50),
    ->     job_title VARCHAR(50),
    ->     salary DECIMAL(10,2),
    ->     hire_date DATE,
    ->     manager_id INT,
    ->     city VARCHAR(50)
    -> );
```

```
mysql> CREATE TABLE products (
    ->     product_id INT PRIMARY KEY,
    ->     product_name VARCHAR(100) NOT NULL,
    ->     category VARCHAR(50) NOT NULL,
    ->     price DECIMAL(10,2) CHECK (price >= 0),
    ->     stock_quantity INT CHECK (stock_quantity >= 0),
    ->     supplier VARCHAR(100)
    -> );
Query OK, 0 rows affected (0.010 sec)
```

```
mysql> CREATE TABLE orders (
    ->     order_id INT PRIMARY KEY,
    ->     customer_name VARCHAR(100),
    ->     product_id INT,
    ->     quantity INT,
    ->     order_date DATE,
    ->     total_amount DECIMAL(10,2),
    ->     status VARCHAR(50)
    -> );
Query OK, 0 rows affected (0.014 sec)
```

```
mysql> SELECT * FROM orders;
+----------+---------------+------------+----------+------------+--------------+------------+
| order_id | customer_name | product_id | quantity | order_date | total_amount | status     |
+----------+---------------+------------+----------+------------+--------------+------------+
|        1 | Alice         |          1 |        2 | 2024-01-05 |    150000.00 | Delivered  |
|        2 | Bob           |          2 |        1 | 2024-01-10 |     45000.00 | Cancelled  |
|        3 | Charlie       |          3 |        3 | 2024-02-02 |     15000.00 | Delivered  |
|        4 | David         |          4 |        1 | 2024-02-14 |     12000.00 | Shipped    |
|        5 | Eva           |          5 |        2 | 2024-02-20 |     36000.00 | Processing |
|        6 | Frank         |          6 |        1 | 2024-03-01 |     15000.00 | Delivered  |
|        7 | Grace         |          7 |        2 | 2024-03-05 |     44000.00 | Shipped    |
|        8 | Hannah        |          8 |        4 | 2024-03-10 |     10000.00 | Delivered  |
|        9 | Ian           |          9 |        5 | 2024-03-15 |      7500.00 | Processing |
|       10 | Jack          |         10 |        1 | 2024-03-18 |      4000.00 | Delivered  |
|       11 | Kevin         |         11 |        1 | 2024-02-22 |     30000.00 | Delivered  |
|       12 | Laura         |         12 |        1 | 2024-01-28 |     55000.00 | Processing |
|       13 | Mike          |         13 |        2 | 2024-02-25 |     36000.00 | Delivered  |
|       14 | Nina          |         14 |        1 | 2024-03-20 |      9000.00 | Shipped    |
|       15 | Oscar         |         15 |        3 | 2024-03-25 |     21000.00 | Delivered  |
+----------+---------------+------------+----------+------------+--------------+------------+
15 rows in set (0.002 sec)

mysql> SELECT * FROM employees;
+-------------+------------+-----------+---------------------------+------------+-----------+----------+------------+------------+---------------+
| employee_id | first_name | last_name | email                     | department | job_title | salary   | hire_date  | manager_id | city          |
+-------------+------------+-----------+---------------------------+------------+-----------+----------+------------+------------+---------------+
|           1 | John       | Watson    | john.watson@company.com   | IT         | Developer | 90000.00 | 2020-02-15 |       NULL | San Francisco |
|           2 | Jane       | Wilson    | jane.wilson@company.com   | IT         | Developer | 85000.00 | 2021-03-10 |          1 | San Francisco |
|           3 | Mark       | Taylor    | mark.taylor@company.com   | IT         | Manager   | 95000.00 | 2019-07-01 |       NULL | San Francisco |
|           4 | Sara       | Johnson   | sara.johnson@company.com  | IT         | Analyst   | 78000.00 | 2021-06-18 |          3 | New York      |
|           5 | Alex       | Brown     | alex.brown@company.com    | IT         | Support   | 60000.00 | 2020-09-12 |          3 | Chicago       |
|           6 | Emily      | Davis     | emily.davis@company.com   | Sales      | Executive | 72000.00 | 2021-11-01 |       NULL | Boston        |
|           7 | Daniel     | Wilson    | daniel.wilson@company.com | Sales      | Associate | 54000.00 | 2022-01-15 |          6 | Chicago       |
|           8 | James      | Anderson  | james.anderson@company.com | Sales      | Associate | 52000.00 | 2022-04-10 |          6 | Boston        |
|           9 | Karen      | Thomas    | karen.thomas@company.com  | Marketing  | Manager   | 88000.00 | 2019-05-20 |       NULL | New York      |
|          10 | Larry      | Martin    | larry.martin@company.com  | Marketing  | Executive | 65000.00 | 2020-08-25 |          9 | Boston        |
|          11 | Robert     | White     | robert.white@company.com  | HR         | Manager   | 70000.00 | 2020-01-05 |       NULL | Chicago       |
|          12 | Nancy      | Harris    | nancy.harris@company.com  | HR         | Executive | 58000.00 | 2021-02-14 |         11 | Boston        |
|          13 | Jacob      | Nelson    | jacob.nelson@company.com  | Finance    | Analyst   | 75000.00 | 2020-10-01 |       NULL | New York      |
|          14 | Aaron      | Clark     | aaron.clark@company.com   | Finance    | Clerk     | 56000.00 | 2021-07-19 |         13 | Chicago       |
|          15 | Megan      | Peterson  | megan.peterson@company.com | Finance    | Clerk     | 53000.00 | 2022-03-11 |         13 | Boston        |
+-------------+------------+-----------+---------------------------+------------+-----------+----------+------------+------------+---------------+
15 rows in set (0.003 sec)
```

```
mysql> SELECT * FROM products;
+------------+-----------------+-------------+----------+----------------+----------+
| product_id | product_name    | category    | price    | stock_quantity | supplier |
+------------+-----------------+-------------+----------+----------------+----------+
|          1 | Laptop          | Electronics | 75000.00 |             20 | Dell     |
|          2 | Smartphone      | Electronics | 45000.00 |             35 | Samsung  |
|          3 | Headphones      | Electronics |  5000.00 |             50 | Sony     |
|          4 | Office Chair    | Furniture   | 12000.00 |             15 | Ikea     |
|          5 | Desk            | Furniture   | 18000.00 |             10 | Ikea     |
|          6 | Printer         | Electronics | 15000.00 |             12 | HP       |
|          7 | Monitor         | Electronics | 22000.00 |             18 | LG       |
|          8 | Keyboard        | Accessories |  2500.00 |             60 | Logitech |
|          9 | Mouse           | Accessories |  1500.00 |             80 | Logitech |
|         10 | Router          | Networking  |  4000.00 |             25 | TP-Link  |
|         11 | Coffee Machine  | Appliances  | 30000.00 |              8 | Philips  |
|         12 | Air Conditioner | Appliances  | 55000.00 |              5 | Voltas   |
|         13 | Water Purifier  | Appliances  | 18000.00 |              9 | Kent     |
|         14 | UPS             | Electronics |  9000.00 |             14 | APC      |
|         15 | Webcam          | Accessories |  7000.00 |              0 | Logitech |
+------------+-----------------+-------------+----------+----------------+----------+
15 rows in set (0.002 sec)
```

```
mysql> --  question 1: Get the 2nd highest salaried person from the Finance department
Query OK, 0 rows affected (0.000 sec)

mysql> SELECT first_name, last_name, salary
    -> FROM employees
    -> WHERE department = 'Finance'   -- selecting only Finance department employees
    -> ORDER BY salary DESC
    -> LIMIT 1 OFFSET 1;
+------------+-----------+----------+
| first_name | last_name | salary   |
+------------+-----------+----------+
| Aaron      | Clark     | 56000.00 |
+------------+-----------+----------+
1 row in set (0.001 sec)

mysql> -- Question 2: Fetch employees from Production department hired between 1998 and 2000
Query OK, 0 rows affected (0.000 sec)

mysql> SELECT *
    -> FROM employees
    -> WHERE department = 'Production'
    ->   AND hire_date BETWEEN '1998-01-01' AND '2000-12-31';
+-------------+------------+-----------+-------------------------+------------+------------+----------+------------+------------+-----------+
| employee_id | first_name | last_name | email                   | department | job_title  | salary   | hire_date  | manager_id | city      |
+-------------+------------+-----------+-------------------------+------------+------------+----------+------------+------------+-----------+
|          16 | Raj        | Kumar     | raj.kumar@company.com   | Production | Supervisor | 80000.00 | 1999-05-10 |       NULL | Chennai   |
|          17 | Anil       | Sharma    | anil.sharma@company.com | Production | Engineer   | 75000.00 | 2000-03-15 |         16 | Pune      |
|          18 | Vijay      | Rao       | vijay.rao@company.com   | Production | Technician | 70000.00 | 1998-08-20 |         16 | Hyderabad |
+-------------+------------+-----------+-------------------------+------------+------------+----------+------------+------------+-----------+
3 rows in set (0.002 sec)
```

```
mysql> -- Question 3: Find the 3rd highest salaried person in Production department
Query OK, 0 rows affected (0.000 sec)

mysql> SELECT first_name, last_name, salary
    -> FROM employees e1
    -> WHERE department = 'Production'
    ->   AND 2 = (
    ->         SELECT COUNT(DISTINCT e2.salary)
    ->         FROM employees e2
    ->         WHERE e2.department = 'Production'
    ->           AND e2.salary > e1.salary
    ->       );
+------------+-----------+----------+
| first_name | last_name | salary   |
+------------+-----------+----------+
| Vijay      | Rao       | 70000.00 |
+------------+-----------+----------+
1 row in set (0.002 sec)

mysql> -- Question 4: Complete report showing how many employees joined in each year
Query OK, 0 rows affected (0.000 sec)

mysql> SELECT
    ->     YEAR(hire_date) AS joining_year,
    ->     COUNT(*) AS total_employees
    -> FROM employees
    -> GROUP BY YEAR(hire_date)
    -> ORDER BY joining_year;
+--------------+-----------------+
| joining_year | total_employees |
+--------------+-----------------+
|         1998 |               1 |
|         1999 |               1 |
|         2000 |               1 |
|         2019 |               2 |
|         2020 |               5 |
|         2021 |               5 |
|         2022 |               3 |
+--------------+-----------------+
7 rows in set (0.003 sec)
```

```
mysql> -- Question 5: Fetch the year in which the maximum number of employees joined
Query OK, 0 rows affected (0.000 sec)

mysql> SELECT
    ->     YEAR(hire_date) AS joining_year,
    ->     COUNT(*) AS total_employees
    -> FROM employees
    -> GROUP BY YEAR(hire_date)
    -> ORDER BY total_employees DESC
    -> LIMIT 1;
+--------------+-----------------+
| joining_year | total_employees |
+--------------+-----------------+
|         2020 |               5 |
+--------------+-----------------+
1 row in set (0.002 sec)

mysql> -- Question 6: Report showing total salary cost per year for each department
Query OK, 0 rows affected (0.000 sec)

mysql> SELECT
    ->     department,
    ->     YEAR(hire_date) AS year,
    ->     SUM(salary) AS total_cost
    -> FROM employees
    -> GROUP BY department, YEAR(hire_date)
    -> ORDER BY department, year;
+------------+------+------------+
| department | year | total_cost |
+------------+------+------------+
| Finance    | 2020 |   75000.00 |
| Finance    | 2021 |   56000.00 |
| Finance    | 2022 |   53000.00 |
| HR         | 2020 |   70000.00 |
| HR         | 2021 |   58000.00 |
| IT         | 2019 |   95000.00 |
| IT         | 2020 |  150000.00 |
| IT         | 2021 |  163000.00 |
| Marketing  | 2019 |   88000.00 |
| Marketing  | 2020 |   65000.00 |
| Production | 1998 |   70000.00 |
| Production | 1999 |   80000.00 |
| Production | 2000 |   75000.00 |
| Sales      | 2021 |   72000.00 |
| Sales      | 2022 |  106000.00 |
+------------+------+------------+
15 rows in set (0.003 sec)
```

```
mysql> -- Question 7: Find the year in which the highest number of female employees joined
Query OK, 0 rows affected (0.000 sec)

mysql> SELECT
    ->     YEAR(hire_date) AS joining_year,
    ->     COUNT(*) AS female_count
    -> FROM employees
    -> WHERE gender = 'Female'
    -> GROUP BY YEAR(hire_date)
    -> ORDER BY female_count DESC
    -> LIMIT 1;
+--------------+--------------+
| joining_year | female_count |
+--------------+--------------+
|         2021 |            4 |
+--------------+--------------+
1 row in set (0.002 sec)

mysql> -- Question 8: Hire date of the current manager of the Human Resources department
Query OK, 0 rows affected (0.000 sec)

mysql> SELECT hire_date
    -> FROM employees
    -> WHERE department = 'HR'
    ->   AND job_title = 'Manager';
+------------+
| hire_date  |
+------------+
| 2020-01-05 |
+------------+
1 row in set (0.002 sec)
```

```
mysql> -- Question 9: Salary of the current manager of the Finance department
Query OK, 0 rows affected (0.000 sec)

mysql> SELECT salary
    -> FROM employees
    -> WHERE department = 'Finance'
    ->   AND job_title = 'Manager';
+----------+
| salary   |
+----------+
| 90000.00 |
+----------+
1 row in set (0.002 sec)

mysql> -- Question 10: Fetch first name and employee number of current department managers
Query OK, 0 rows affected (0.000 sec)

mysql> SELECT
    ->     employee_id,
    ->     first_name
    -> FROM employees
    -> WHERE job_title = 'Manager';
+-------------+------------+
| employee_id | first_name |
+-------------+------------+
|           3 | Mark       |
|           9 | Karen      |
|          11 | Robert     |
|          19 | Peter      |
+-------------+------------+
4 rows in set (0.002 sec)
```